

Spatial Statistics - Research

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1 A General Periodic Fourier–State-Space Spatio–Temporal Model

1.1 Model setup

We consider a real-valued Gaussian spatio–temporal process

$$\{X_t(x) : x \in [0, 2\pi], t \in \mathbb{Z}\},$$

defined on a one-dimensional spatial domain with periodic boundary conditions (a circle) and discrete time. For each time t , the spatial field $X_t(\cdot)$ is represented by a truncated Fourier series

$$X_t(x) = a_0(t) + \sum_{j=1}^J \left[a_j(t) \cos(jx) + b_j(t) \sin(jx) \right], \quad x \in [0, 2\pi], \quad (1)$$

where J is a truncation level and

$$a_j(t), b_j(t), \quad j = 0, \dots, J, t \in \mathbb{Z},$$

are jointly Gaussian random coefficients. The representation (1) is exact in the limit $J \rightarrow \infty$ and is a finite-dimensional approximation in practice.

For $j \geq 1$, the pair $(a_j(t), b_j(t))$ corresponds to the j th spatial frequency (wavenumber). Low j terms encode large-scale, slowly varying spatial structure, while high j terms encode fine-scale spatial detail.

1.2 Spatial margin: Matérn-like variances

The spatial covariance at each fixed time is controlled by a Matérn-like spectral density in the Fourier domain. For $j = 0, \dots, J$ we specify

$$\sigma_j^2 \propto (\kappa^2 + j^2)^{-(\nu+1/2)}, \quad \kappa = \frac{\sqrt{2\nu}}{\ell}, \quad (2)$$

where

- $\nu > 0$ is the Matérn smoothness parameter,
- $\ell > 0$ is a spatial range parameter,
- $\kappa = \sqrt{2\nu}/\ell$ is an inverse scale parameter.

On $\mathbb{R}$, the Matérn covariance with parameters (ν, ℓ) has spectral density proportional to $(\kappa^2 + \omega^2)^{-(\nu+1/2)}$ in one spatial dimension, and sample paths that are $\lceil \nu \rceil - 1$ times mean-square differentiable. The discrete Fourier variances σ_j^2 in (2) therefore give a Matérn-like spatial margin on the periodic domain.

For simulation it is convenient to normalize these variances so that the total marginal variance of $X_t(x)$ is a chosen constant σ_{tot}^2 . Defining

$$\tilde{\sigma}_j^2 = (\kappa^2 + j^2)^{-(\nu+1/2)},$$

we set

$$\sigma_j^2 = \sigma_{\text{tot}}^2 \cdot \frac{\tilde{\sigma}_j^2}{\sum_{k=0}^J \tilde{\sigma}_k^2}, \quad j = 0, \dots, J. \quad (3)$$

1.3 Temporal dynamics: block-diagonal AR(1) with rotation and correlated innovations

We collect the Fourier coefficients into a state vector

$$\mathbf{Z}_t = (a_0(t), a_1(t), \dots, a_J(t), b_1(t), \dots, b_J(t))^\top \in \mathbb{R}^{2J+1}. \quad (4)$$

The temporal evolution is governed by a *block-diagonal* linear Gaussian state-space model. For the constant mode $j = 0$, we have a scalar AR(1):

$$a_0(t+1) = \rho_0 a_0(t) + \eta_{0,t}, \quad \eta_{0,t} \sim \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2)), \quad (5)$$

with $|\rho_0| < 1$ ensuring stationarity.

For each mode $j \geq 1$, the cosine and sine coefficients evolve jointly according to a 2×2 block AR(1):

$$\begin{pmatrix} a_j(t+1) \\ b_j(t+1) \end{pmatrix} = M_j \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} + \begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix}, \quad (6)$$

where the 2×2 transition matrix is

$$M_j = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix}, \quad (7)$$

with

- $\rho_j \in (0, 1)$: amplitude persistence (decay rate),
- $\theta_j \in \mathbb{R}$: rotation angle (controls temporal mixing of a_j and b_j).

The 2×2 innovation covariance for mode j is

$$\begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\varepsilon,j}), \quad \Sigma_{\varepsilon,j} = \sigma_j^2(1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}, \quad (8)$$

where $\phi_j \in (-1, 1)$ is the *within-frequency correlation parameter* between the cosine and sine innovations. For $\Sigma_{\varepsilon,j}$ to be positive definite, we require $|\phi_j| < 1$.

The innovations are independent across different frequencies j and across time t .

Physical interpretation

Rotation parameter θ_j :

- When $\theta_j = 0$, the transition matrix $M_j = \rho_j I_2$, so $a_j(t+1)$ depends only on $a_j(t)$ and $b_j(t+1)$ depends only on $b_j(t)$ (no temporal mixing).
- When $\theta_j \neq 0$, we have

$$a_j(t+1) = \rho_j[\cos \theta_j \cdot a_j(t) - \sin \theta_j \cdot b_j(t)] + \eta_{a,j,t},$$

$$b_j(t+1) = \rho_j[\sin \theta_j \cdot a_j(t) + \cos \theta_j \cdot b_j(t)] + \eta_{b,j,t},$$

so $a_j(t+1)$ depends on *both* $a_j(t)$ and $b_j(t)$. This induces *phase rotation*: spatial patterns propagate or rotate around the circle over time.

- The eigenvalues of M_j are $\rho_j e^{\pm i\theta_j}$, so the spectral radius is $|\rho_j| < 1$ regardless of θ_j , ensuring stationarity.

Innovation correlation parameter ϕ_j :

- When $\phi_j = 0$, the innovations $\eta_{a,j,t}$ and $\eta_{b,j,t}$ are independent, so $a_j(t)$ and $b_j(t)$ are uncorrelated at the same time t (at stationarity).
- When $\phi_j \neq 0$, the innovations are correlated: $\text{Cov}(\eta_{a,j,t}, \eta_{b,j,t}) = \sigma_j^2(1 - \rho_j^2)\phi_j$. At stationarity, this induces

$$\text{Cov}(a_j(t), b_j(t)) \neq 0.$$

This creates *instantaneous correlation* between the cosine and sine components at the same time.

- Typical choice: $\phi_j = \phi_0/(1+j)$ or $\phi_j = \phi_0 e^{-\gamma j}$, so that lower frequencies have stronger instantaneous correlation than higher frequencies.

Combined effect:

- $\theta_j \neq 0$ creates *temporal cross-dependence*: $a_j(t)$ depends on $b_j(t-1)$ and vice versa.
- $\phi_j \neq 0$ creates *contemporaneous correlation*: $a_j(t)$ and $b_j(t)$ are correlated at the same time.
- Together, they allow rich within-frequency dynamics while maintaining the block-diagonal structure (frequencies still evolve independently of each other).

1.4 Stationary covariance of $(a_j(t), b_j(t))$ via Lyapunov equation

At stationarity, the 2×2 covariance matrix

$$\Sigma_j = \text{Cov} \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix}$$

satisfies the discrete-time Lyapunov equation

$$\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}. \quad (9)$$

For the specific structure (7)–(8), the solution is

$$\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}. \quad (10)$$

This can be verified by direct substitution: since M_j is a rotation-scaling matrix with $M_j M_j^\top = \rho_j^2 I_2$, we have

$$M_j \Sigma_j M_j^\top = \rho_j^2 \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix},$$

and thus

$$M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j} = \rho_j^2 \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix} + \sigma_j^2 (1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix} = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix} = \Sigma_j,$$

confirming (10).

Therefore, at stationarity:

$$\text{Var}(a_j(t)) = \sigma_j^2, \quad (11)$$

$$\text{Var}(b_j(t)) = \sigma_j^2, \quad (12)$$

$$\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j. \quad (13)$$

Note that the stationary covariance (10) depends on ϕ_j but *not* on θ_j . The rotation angle θ_j affects the temporal evolution and cross-lag covariances, but not the marginal variances or contemporaneous correlation at a fixed time t .

1.5 Nonseparability via frequency-dependent persistence

The key nonseparable feature arises from allowing the AR(1) persistence parameter ρ_j to depend on the spatial frequency j . We propose an *algebraic (power-law) decay* parameterization

$$\rho_j = \frac{1}{(1 + \lambda j^\alpha)^\beta}, \quad j = 1, \dots, J, \quad (14)$$

with $\rho_0 \in (0, 1)$ set separately, and parameters

- $\lambda > 0$: controls the overall decay strength across frequencies,

- $\alpha > 0$: controls the frequency dependence of the power,
- $\beta > 0$: controls the decay exponent.

For large j , this decays as $\rho_j \sim j^{-\alpha\beta}$, which is an *algebraic tail* in contrast to the exponential tail of the form $\exp(-\lambda j^\alpha)$. Algebraic decay is slower than exponential decay and is common in systems with long-range temporal dependence (e.g. turbulence, climate, neural signals). The parameter β provides fine control over the decay rate: smaller β gives slower decay and longer temporal memory even at high frequencies.

Under this structure, low-frequency modes (small j) have $\rho_j \approx 1$ and decorrelate slowly in time, while high-frequency modes have smaller ρ_j and decorrelate more rapidly. Consequently, the effective spatial correlation structure changes with time lag: at short time lags, both low and high spatial frequencies contribute, whereas at long time lags, only low-frequency components remain. This space–time covariance cannot be written as a product $C_{\text{space}}(h)C_{\text{time}}(k)$ and is therefore nonseparable.

1.6 Space–time covariance

Under the block-diagonal structure with rotation and correlated innovations, the space–time covariance

$$C(h, k) = \mathbb{E}[X_t(x)X_{t+k}(x+h)]$$

can be computed when the different modes are uncorrelated across frequencies. For a zero-mean process with the dynamics (5)–(6), and using the fact that

$$\mathbb{E}[a_j(t)a_j(t+k)] = \mathbb{E}[b_j(t)b_j(t+k)] = \sigma_j^2 \rho_j^{|k|},$$

we obtain

$$C(h, k) = \sum_{j=0}^J \sigma_j^2 \rho_j^{|k|} \cos(jh), \quad (15)$$

where $\cos(0 \cdot h) = 1$ for the $j = 0$ term.

Derivation sketch: Substituting (1) and using linearity of expectation, cross terms between different modes vanish. For a fixed mode j , the contribution is

$$\mathbb{E}[a_j(t)a_j(t+k)] \cos(jx) \cos(j(x+h)) + \mathbb{E}[b_j(t)b_j(t+k)] \sin(jx) \sin(j(x+h)).$$

Using the trigonometric identity $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = jx$, $B = j(x+h)$, this simplifies to

$$\mathbb{E}[a_j(t)a_j(t+k)] \cos(jh).$$

For a stationary AR(1) with persistence ρ_j and marginal variance σ_j^2 , we have $\mathbb{E}[a_j(t)a_j(t+k)] = \sigma_j^2 \rho_j^{|k|}$, yielding (15).

Note: The contemporaneous correlation $\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j$ does not appear in (15) because the cross terms $\mathbb{E}[a_j(t)b_j(t+k)] \cos(jx) \sin(j(x+h))$ integrate to zero over the periodic domain when computing the covariance as a function of (h, k) alone. The

parameter ϕ_j affects the joint distribution of $(a_j(t), b_j(t))$ and the temporal cross-covariances $\mathbb{E}[a_j(t)b_j(t+k)]$ for $k \neq 0$, but does not change the space–time covariance formula at integer lags.

Similarly, the rotation parameter θ_j creates phase drift and affects the realizations, but does not change the stationary marginal variances or the form of (15).

1.7 Parameter overview

The full model includes the following parameters:

- **Fourier truncation:** J (integer, $J \geq 0$). Controls the finest spatial scale represented. Larger J allows finer spatial detail; in practice J is chosen so that the total variance in modes $j > J$ is negligible.
- **Matérn spatial parameters:**
 - $\nu > 0$ (smoothness): larger ν gives smoother spatial fields. Sample paths are $\lceil \nu \rceil - 1$ times mean-square differentiable.
 - $\ell > 0$ (range): larger ℓ increases the spatial correlation length by shifting spectral mass to lower frequencies.
 - $\sigma_{\text{tot}}^2 > 0$ (total marginal variance): overall scale of the field.
 - $\kappa = \sqrt{2\nu}/\ell$ (inverse range): internal scale parameter used in the spectral envelope (2).
- **Temporal persistence parameters:**
 - $\rho_0 \in (0, 1)$: AR(1) persistence for the mean mode $a_0(t)$, set separately to avoid nonstationarity.
 - $\rho_j \in (0, 1)$, $j = 1, \dots, J$, given by (14) with
 - * $\lambda > 0$: controls overall decay speed across frequencies,
 - * $\alpha > 0$: controls how strongly persistence changes with j ,
 - * $\beta > 0$: decay exponent; smaller β gives slower (longer-memory) decay.
- **Phase rotation parameters:**
 - $\theta_j \in \mathbb{R}$, $j = 1, \dots, J$: rotation angle for mode j in the 2×2 block M_j (equation (7)).
 - $\theta_j = 0$ recovers the no-rotation model (cosine and sine evolve independently over time).
 - Typical parameterization: $\theta_j = \theta_0/j$ or $\theta_j = \theta_0/(1+j)$, so that lower frequencies rotate more than higher frequencies.
- **Innovation correlation parameters:**

- $\phi_j \in (-1, 1)$, $j = 1, \dots, J$: within-frequency correlation between $\eta_{a,j,t}$ and $\eta_{b,j,t}$ (equation (8)).
- $\phi_j = 0$ gives independent innovations (cosine and sine uncorrelated at the same time).
- $\phi_j \neq 0$ induces $\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j$ at stationarity.
- Typical parameterization: $\phi_j = \phi_0/(1 + j)$ or constant $\phi_j = \phi_0$ for all j .

- **Time horizon:** T (integer, $T \geq 1$). Number of time points simulated or observed.

By setting $\theta_j = 0$ and $\phi_j = 0$ for all j , the model reduces to the simple independent-mode AR(1) with uncorrelated cosine and sine components. Nonzero θ_j and ϕ_j introduce richer within-frequency dynamics: temporal mixing and phase rotation (θ_j), and instantaneous correlation (ϕ_j).

1.8 Simulation algorithm

We now describe the simulation algorithm step by step.

Step 1: choose parameters

Select values for

$$(J, T, \nu, \ell, \sigma_{\text{tot}}^2, \lambda, \alpha, \beta, \rho_0, \{\theta_j\}_{j=1}^J, \{\phi_j\}_{j=1}^J).$$

Step 2: construct the Matérn-like spectrum

Compute the unnormalized spectral weights

$$\tilde{\sigma}_j^2 = (\kappa^2 + j^2)^{-(\nu+1/2)}, \quad j = 0, \dots, J,$$

with $\kappa = \sqrt{2\nu}/\ell$, then normalize as in (3) to obtain σ_j^2 such that $\sum_{j=0}^J \sigma_j^2 = \sigma_{\text{tot}}^2$.

Step 3: compute mode-specific AR(1) coefficients

Compute the AR(1) persistence parameters ρ_j for $j = 1, \dots, J$ using (14), and set ρ_0 by hand.

Step 4: build the 2×2 transition matrices

For each $j = 1, \dots, J$, construct

$$M_j = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix}.$$

Step 5: build the 2×2 innovation covariances

For mode $j = 0$:

$$\text{Var}(\eta_{0,t}) = \sigma_0^2(1 - \rho_0^2).$$

For each mode $j \geq 1$, construct

$$\Sigma_{\varepsilon,j} = \sigma_j^2(1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}.$$

Verify that $|\phi_j| < 1$ to ensure $\Sigma_{\varepsilon,j}$ is positive definite.

Step 6: initialize the Fourier coefficients

Initialize the process at $t = 1$ from the stationary distribution:

$$a_0(1) \sim \mathcal{N}(0, \sigma_0^2),$$

and for $j = 1, \dots, J$

$$\begin{pmatrix} a_j(1) \\ b_j(1) \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}\right),$$

independently across j .

Step 7: simulate the AR(1) dynamics

For time $t = 1, \dots, T - 1$:

1. Update the constant mode using (5):

$$a_0(t+1) = \rho_0 a_0(t) + \eta_{0,t}, \quad \eta_{0,t} \sim \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2)).$$

2. For each $j = 1, \dots, J$, update using (6):

$$\begin{pmatrix} a_j(t+1) \\ b_j(t+1) \end{pmatrix} = M_j \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} + \begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix},$$

where

$$\begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \Sigma_{\varepsilon,j}\right),$$

independent across j and across t .

This yields arrays

$$\{a_j(t)\}_{t=1,\dots,T}^{j=0,\dots,J} \quad \text{and} \quad \{b_j(t)\}_{t=1,\dots,T}^{j=1,\dots,J}.$$

Step 8: reconstruct the spatial field on a grid

To visualize or analyze the field, choose a spatial grid

$$x_1, \dots, x_N \in [0, 2\pi],$$

for example $x_n = 2\pi(n-1)/(N-1)$, and compute

$$X_t(x_n) = a_0(t) + \sum_{j=1}^J [a_j(t) \cos(jx_n) + b_j(t) \sin(jx_n)],$$

for all $t = 1, \dots, T$ and $n = 1, \dots, N$. This produces a $T \times N$ matrix X that can be plotted as time series, spatial snapshots, or as an image.